API TITLE*1993 INT ** 0732290 0534167 637

1993 Interpretations—Technical Publications

Answers to Technical Inquiries

AUGUST 1994

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



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API TITLE*1993 INT ** 0732290 0534168 573

1993 Interpretations—Technical Publications

Answers to Technical Inquiries

Industry Operations Group

AUGUST 1994

American Petroleum Institute

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FOREWORD

This publication is a compilation of interpretations issued in 1993 by the Industry Operattions Group of the American Petroleum Institute. API has issued these interpretations in response to questions concerning the meaning of its standards and in consideration of suggestions for revisions to its standards.

API welcomes questions, suggestions, and comments concerning its standards. However, API can only respond to questions concerning application of its standards that are stated clearly and that strictly pertain to the meaning of one of its standards. API staff will make interpretations of standards in writing after consultation, if necessary, with the appropriate committee officers and the committee membership.

Suggestions for revision to an API standard should include supporting technical information. Inquiries that do not provide the information needed for complete understanding will be returned.

As a matter of policy, API does not approve, certify, rate, or endorse any item, contribution, proprietary device, or activity. Moreover, API does not act as a consultant on specific engineering problems or on the general understanding or application of its standards. Therefore, inquiries requesting such considerations will be returned.

API does not normally respond to questions seeking the rationale for requirements in its standards. These requirements are based upon consideration of technical data and the judgment and skill of experienced engineering and technical personnel, representing both users and manufacturers, who serve on API standards committees.

API committees meet regularly to consider written requests for interpretations and revisions of standards, as well as to develop new standards as dictated by technological development. All technical meetings during which requirements of API standards are considered are open to the public. Notices of these meetings are published in *Standards Action*, a publication of the American National Standards Institute.* A report of standardization activities is also available from individual API departments.

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HOW TO PREPARE A TECHNICAL INQUIRY

Inquiries sent to the American Petroleum Institute should either request an interpretation of an API standard or ask that a revision to an API standard be considered on the basis of new data or technology. The following guidelines are recommended for submitting inquiries:

1. Format: Include your name and mailing address on all correspondence. Your letter should be typewritten (although legible handwritten letters will be considered), and mailed to:

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005

2. Scope: Limit your inquiry to a single subject or closely related subjects. An inquiry letter concerning unrelated subjects will be returned.

3. Background: State the purpose of your inquiry and your understanding of the standard's requirements. Provide all information needed for a complete understanding of your inquiry. Refer to the applicable edition, revision, paragraphs, figures, and tables. Include sketches, if needed.

4. Inquiry: State your inquiry in a condensed, precise question format, omitting any superfluous background information. The question itself should be technically and editorially correct. Try to compose your question in such a way that "yes" or "no," with provisos if needed, would be a suitable reply. For example, you could ask whether or not your understanding of a standard's requirements is correct.

5. Suggestion for a Revision: When proposing a revision to an API standard, you should include supportive technical reasons and data, along with proposed new wording. Comments made concerning the technical basis for a requirement of an API standard should include relevant notes and suggestions. These suggestions and comments will be forwarded to the appropriate API subcommittee for the next review of that standard.

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API TITLE*1993 INT ** ■ 0732290 0534173 930 ■

Issue Date: March 31, 1993

Publication: API Specification 5CT, *Specification for Casing and Tubing*

Edition/Date: Fourth Edition, November 1, 1992
with Errata, November 1, 1992

Question 1: Under the section on "Outside Diameter" (OD), the statement is made to the effect, "Measurements shall be made with calipers or snap gauges." Does this requirement apply only to upset casing or to all OD measurements of tubing and casing?

Reply 1: All OD measurements of tubing and casing.

Question 2: Is a "pie tape" acceptable for an API licensee to utilize to perform OD measurements to verify casing and/or tubing to meet API Specification 5CT requirements?

Reply 2: No. I interpret the references to calipers or snap-gages in Table 6.7 of Specification 5CT to exclude the use of diameter tapes. Note that diameter tapes are referenced in Table 6.3 of Specification 5L.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534174 877 ■

Issue Date: May 3, 1993

Publication: API Specification 5CT, *Specification for Casing and Tubing*

Edition/Date: Fourth Edition, November 1, 1992
with Errata, November 1, 1992

Question: In lieu of API Specification 5CT, Fourth Edition, November 1, 1992, Appendix D, D.1, D.3, and D.6, pertaining to the paint stencil markings of pipe, can the actual length and weight of the entire pipe be stenciled at the beginning of the sequence of the stenciled message?

Reply: Yes, I interpret the actual length and weight to be examples of "additional markings" allowed by D.1, provided all other requirements (such as, the location specified in D.6) are met.

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API TITLE*1993 INT ** ■ 0732290 0534175 703 ■

Issue Date: December 13, 1993

Publication: API Specification 5CT, *Specification for Casing and Tubing*

Edition/Date: Fourth Edition, November 1, 1992
with Errata, November 1, 1992

Question: Does the reference to a *threader* in 6.9 of Specification 5CT require a drift test in addition to the drift test required by the pipe manufacturer?

Reply: The reference to a *threader* in API Specification 5CT is defined in 11.3 as follows:

"A pipe threader shall operate one or more threading machines capable of threading pipe ends to the dimensions and tolerances specified in API Standard 5B."

All Standard 5B connections require drift testing in accordance with 6.9 of Specification 5CT in addition to the full length drift test. I am aware of cases where pipe has failed the drift test after threading and coupling.

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API TITLE*1993 INT ** ■ 0732290 0534176 64T ■

Issue Date: January 21, 1993

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question: Does the stated outside diameter tolerance of line pipe larger than size 20 per Table 6.3 of the API Specification 5L mean that pipe ends measuring 1/32 inch smaller (or 0.031 inch) or 3/32 inch larger (or 0.094 inch) than specified by 0.001 inch as measured with a diameter tape graduated in thousandths would be rejectable?

Reply: Yes, provided the accuracy of this measuring instrument does not exceed this amount.

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API TITLE*1993 INT ** ■ 0732290 0534177 586 ■

Issue Date: February 26, 1993

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question 1: Is API Specification 5L, Thirty-Seventh Edition, the specification that API 5L line pipe is required to be manufactured to?

Reply 1: The thirty-seventh edition of API Specification 5L became effective on May 31, 1988. The thirty-eighth edition of API Specification 5L became effective on May 1, 1990. During that period the requirements of the thirty-seventh edition were in effect for API licensed manufacturers using the monogram on of API Specification 5L products.

Question 2: Does the National Bureau of Standards Certificate of Analysis Standard Reference Material 1269, *Line Pipe Steel,* reflect the required API 5L chemistry for ERW Non Expanded API 5L Grade B line pipe?

Reply 2: No. This is the chemical analysis of one standard reference material which may be used for calibration of test equipment per Section 3 of of API Specification 5L.

Question 3: Does the addition of Aluminum which was added for deoxidation pertain to 3.1, "The addition of such otherwise desirable elements may alter the weldability of the pipe and caution should be exercised in the determination of the quantity which may be added for any particular size and thickness of pipe", or does the addition of Aluminum pertain to 3.2, Heat Analysis, "If alloys other than those specified in Table 3.1 for a particular grade are added for other than deoxidation purposes, the heat analyses, including the alloy additions, shall be reported for each heat applied to the purchaser's order?"

Reply 3: Paragraph 3.1.

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API TITLE*1993 INT ** 0732290 0534178 412

Issue Date: March 9, 1993

Publication: API Specification 5L, *Specification for Line Pipe*

Edition/Date: Fortieth Edition, November 1, 1992

Question: API Specification 5L, Fortieth Edition, has moved to dimensionless quantities for size and weight designations. Consequently, 11.1.c has been modified and, logically, the reference to marking size and weight in metric units has been deleted. In order to recognize this possibility, could API Specification 5Lbe modified to include API Specification 5CT wording regarding additional markings and compatible standards?

Reply: I agree that revisions to the Fortieth Edition of API Specification 5L were not intended to prevent the marking of size and weight in metric units. In order to clarify this point, the next edition of Specification 5L, 11.1 (new Subparagraph d), and Appendix H.1 (new Subparagraph d), will be revised per resolution of API C5 Ballot Item 2247 as follows:

"Additional markings including those for compatible standards following the specification marking are allowed and may be applied as desired by the manufacturer or as requested by the purchaser."

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API TITLE*1993 INT ** ■ 0732290 0534179 359 ■

Issue Date: May 14, 1993

Publication: Specification 6A, *Specification for Wellhead and Christmas Tree Equipment*

Edition/Date: Sixteenth Edition, October 1, 1989
with Supplement 2, July 1, 1992

Question: When imperfections are removed from weld overlay and machining extends into the base metal to remove HAZ and over-sizing area from original pre-clad dimensions does this create a repair weld to the base metal?

Reply: No. Removal of defects in weld overlay weldments followed by re-welding is an accepted practice. Removal of the weldment HAZ would not create a repair to the base metal.

Question: Are there specific limitations as to how much base metal may be removed when removing imperfections from the overlay weld prior to the reapplication of overlay?

Reply: No.

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API TITLE*1993 INT ** 0732290 0534180 070

Issue Date: October 15, 1993

Publication: Specification 6A, *Specification for Wellhead and Christmas Tree Equipment*

Edition/Date: Sixteenth Edition, October 1, 1989
with Supplement 2, July 1, 1992

Question 1: In Table 302.3 there is no mention of lock screws. Is it correct that lock screws have the same minimum material requirements as stems?

Reply 1: No. Table 302.3 does not currently address the material requirements for lock screws, leaving them subject to manufacturer specification (reference 906.4b and Section 917).

Question 2: Is it acceptable to manufacture the "parent bowl" or "false bowl" (for dual tubing hangers) from low alloy non-NACE materials (when the mandrel hangers are for EE or FF service)?

Reply 2: If the tubing spool in which the "parent bowl," "false bowl," or "parent hanger" rests is rated for AA or BB service, the answer is YES. If the tubing spool in which the "parent bowl," "false bowl," or "parent hanger" rests is rated for a material class other than AA or BB, the answer is NO.

The parent hanger is exposed to the same environment as the tubing spool in which it rests and should meet the same material class unless otherwise specified by the end user.

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API TITLE*1993 INT ** ■ 0732290 0534181 T07 ■

Issue Date: October 15, 1993

Publication: Specification 6A, *Specification for Wellhead and Christmas Tree Equipment*

Edition/Date: Sixteenth Edition, October 1, 1989
with Supplement 2, July 1, 1992

Question: Is hardness testing of body weldments required for PSL-1 or PSL-2 equipment with material classes DD, EE, FF, or HH?

Reply: No. Table 605.2 does not invoke hardness testing for PSL-1 or PSL-2 production weldments.

API Specification 6A addresses the capability of a production weldment to meet NACE MR-01-75 requirements by providing extensive controls on the WPS and PQR in Section 500. These up front controls reflect the impossibility of accurately checking hardness throughout the heat affected zone of a production weldment without destroying or cutting up the part.

If deemed necessary for specific applications, API Specification 6A does not prohibit additional requirements, by manufacturer/purchaser agreement, such as sacrificial components for hardness testing of weld heat affected zones.

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API TITLE*1993 INT ** ■ 0732290 0534182 943 ■

Issue Date: December 16, 1993

Publication: Specification 6A, *Specification for Wellhead and Christmas Tree Equipment*

Edition/Date: Sixteenth Edition, October 1, 1989
with Supplement 2, July 1, 1992

Question 1: Reference 503.1b(2). Do the deposited weld metal mechanical properties, as determined by the procedure qualification record (PQR), include all base material properties included in Table 404.1?

Reply 1: Yes. All the mechanical properties that are specified either by API Specification 6A requirements or by the manufacturer would need to be met or exceeded.

Question 2: Reference 605.3b, Volumetric Examination (PSL 3-4). Calibration. How should the 1/16-inch side drilled hole be oriented for the angle beam technique? How should the sound beam intersect the side drilled hole in this examination?

Reply 2: API Specification 6A currently does not specify how the side drilled hole should be oriented or how the sound beam should intersect it.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534183 88T ■

Issue Date: December 16, 1993

Publication: Specification 6A, *Specification for Wellhead and Christmas Tree Equipment*

Edition/Date: Sixteenth Edition, October 1, 1989
with Supplement 2, July 1, 1992

Question: Does 504.2 of API Specification 6A apply to weld repair of raw material defects or only to weld repair of machining defects as is implied by 504.2a which states: "Bolt Hole, Tapped Hole and Machined Blind Hole Repair–Performance Qualification"?

Reply: Paragraph 504.2 applies to repair weldments for PSL 2-3 bodies, bonnets, and end and outlet connections, including those made to correct raw material defects.

Subparagraph 504.2a applies only to "bolt hole, tapped hole and machined blind hole repair". It lists the performance qualification requirements for weld repairs to these holes, regardless of whether the defect is due to a machining defect or a raw material defect. Subparagraph 504.2a does not apply to the repair of a casting defect that is not associated with a bolt hole, tapped hole or machined blind hole.

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API TITLE*1993 INT ** ■ 0732290 0534184 716 ■

Issue Date: October 15, 1993

Publication: Specfication 6FA, *Specification for Fire Test for Valves*

Edition/Date: Second Edition, February 15, 1994

Question: Does a successful fire test on a 6-inch Class 900 ball valve, in which the ball is made of electroless nickel plated carbon steel, also qualify a 6-inch Class 900 ball valve of the same basic design but with a different material for the ball; that is, nodular iron or stainless steel?

Reply: Yes.

Question: Would the above test qualify 10- and 12-inch size Class 900 valves of the same basic design as the tested valve and same materials with respect to seat to closure member seal, seat to body seal, but the ball material being nodular iron?

Reply Yes.

Question: Would the fire tested 6-inch Class 900 valve qualify a 6 in. x 4 in. x 6 in. Class 900 valve because the end connection size is the same, and the basic design remains the same as the fire-tested full bore valve except that the ball material for this 6 in. x 4 in. x 6 in. valve is 316 stainless steel?

Reply: Yes.

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API TITLE*1993 INT ** 0732290 0534185 652

Issue Date: January 13, 1993

Publication: API Specification 7, *Specification for Rotary Drilling Equipment*

Edition/Date: Thirty-Seventh Edition, August 1, 1990

Question: Is it acceptable to stamp a bi-center PDC bit with an API monogram under the marking requirements of API Specification 7 if the outside diameter for the drilling condition meets Table 7.3 tolerances?

Answer: No, bi-center bits are considered special purpose tools, as are jars and under-reamers, that are outside the scope of API Specification 7.

For example, one of the purposes of the outside diameter tolerance is to assist the bit user in selecting interchangeable bits during drilling; however, API Specification 7 does not address how to determine the size of the well bore hold produced by this bit for various formations or conditions.

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Issue Date: March 11, 1993

Publication: Specification 10A, *Specification for Well Cement*

Edition/Date: Twenty-First Edition, September 1, 1991

Question: Please refer to API Specification 10A, Twenty-First Edition, September 1, 1991, Section 10, Packing.

In our company, cement is dispatched in sacks (42.5 kilograms), jumbo bags (1.5 tonne) and in bulk (trucks and ships). For sack dispatches, the sacks are either loaded directly from the packing unit onto trucks (280 sacks each) or directly onto pallets (35 sacks each) and plastic wrapped.

Customers are protected against short-weight by weight checks that are performed and recorded at each packing unit at FREQUENT, SPECIFIED INTERVALS. Corrective action is immediately taken if the weights are found to be out of specification.

Considering what is presently being done, is it still necessary for us to determine the average weight of 5 percent of all sacks in a shipment.

Reply: The specification states that each sack shall be within ±2 percent of the stated sack weight. It also says that you will show that 5 percent of all the sacks in a shipment have the specified weight within these limits. You do not have to weigh 5 percent of all the sacks in a shipment. However, you must have adequate controls and testing procedures in place to be able to show that you are testing 5 percent of your bag cement for this specified weight requirement. If your weight checks at each packaging unit is following a 5 percent sampling procedure then you are in compliance. If not then you are out of compliance with Specification 10A.

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API TITLE*1993 INT ** ■ 0732290 0534187 425 ■

Issue Date: July 16, 1993

Publication: Specification 10A, *Specification for Well Cements*

Edition/Date: Twenty-First Edition, September 1, 1991

Question: Is a certificate of calibration for the following equipment used to test the cement acceptable to the API as evidence of the calibration of the equipment?

1. Atmospheric consistometer
2. Pressurized consistometer
3. Compressive strength tester
4. Dual speed slurry mixer

Reply: Yes, a certificate of calibration is acceptable as evidence that the equipment listed above has been calibrated.

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API TITLE*1993 INT ** 0732290 0534188 361

Issue Date: October 2, 1993

Publication: Specification 10D, *Specification for Bow-Spring Casing Centralizers*

Edition/Date: Fourth Edition, January 1, 1991

Question: Refer to 6.1.d of API Specification 10D, must we apply 3 times the restoring force as Table 3.1 or apply the force in three different places.

Reply: You must apply 3 times the restoring force as is shown in Table 3.1. The force is not applied in three places.

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API TITLE*1993 INT ** [redacted] 0732290 0534189 2T8 [redacted]

Issue Date: September 2, 1993

Publication: RP 13I, *Recommended Practice Standard Procedure for Laboratory Testing Drilling Fluids*

Edition/Date: Fourth Edition, June 1, 1990

Question: Referring to API Recommended Practice 13I (RP 13I), Third Edition, September 1, 1988 and the Fourth Edition, June 1, 1990, and in particular Section 1, Particle Size Analysis for Fines in Barite, 1.2, Equipment, Paragraph (a), there is a variance between the two publications in the first sentence of the paragraph. The third edition quotes 3.6 grams sodium carbonate/1000 cm^3 of solution and the fourth edition quotes 3.6 grams sodium carbonate/100 cm^3 of solution.

Is this a typographical error or a deliberate revision.

Reply: The 100 cm^3 is a typo and should be 1000 cm^3. We will correct it in the next edition of Recommended Practice 13I.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534190 T1T ■

Issue Date: November 17, 1993

Publication: API Chapter 14, "Natural Gas Fluids Measurement," Section 3, "Concentric, Square-Edge Orifice Meters, Part 2, Specification and Installation Requirements" of the *Manual of Petroleum Measurement Standards*

Edition/Date: Third Edition, February 1991

Question 1: Paragraph 2.5.1.4.3. states: "A recess resulting from an orifice plate gasket or sealing device of more than 0.25 inch but less than or equal to 0.5 inch in length, *as measured parallel to pipe axis*, does not require diameter ratio (Br) limitation or additional uncertainty if the depth of the recess is within limitations of 2.5.1.3."

The italicized wording is not part of the specification but is assumed to be implied. Is this assumption correct?

Reply 1: The wording *as measured parallel to pipe axis* is implied as written in your question.

Question 2: Paragraph 2.5.1.4.3 references 2.5.1.3, Tolerances and Restrictions. Subparagraph 2.5.1.3.1.1 references Table 2.5 for determining the allowable tolerance of the orifice plate holder recess on the upstream side of the orifice plate. Is this the correct interpretation of the standard for determining the maximum allowable recess for the orifice plate holder on the upstream side of the orifice plate?

Reply 2: Your interpretation is correct. Notice that Table 2-5 is an example based on Equation 2-3 for the upstream side of the orifice plate.

Question 3: Paragraph 2.5.1.3.2 references downstream roundness and it is interpreted that the downstream recess of the orifice holder is covered by 2.5.1.3.2. Is this correct?

Reply 3: Your interpretation is correct. Notice that Equation 2-5 of 2.5.1.3.2 applies to downstream tolerances.

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API TITLE*1993 INT ** ■ 0732290 0534191 956 ■

Issue Date: November 17, 1993

Publication: *Manual of Petroleum Measurement Standards*

Edition/Date: General Question

Question: The answers to your questions are detailed below:

Reply: There is a correspondence between density at 20°C and 60°F, as shown below. However, the procedure shows the conversion from 60°F values to 20°C densities. In order to do the reverse, as you request, you will have to iterate the calculation. That is, you will have to guess a 60°F value and perform the calculation until you match the 20°C value you want to get.

0.8780 kg/L at 20°C does correspond to 0.8820 relative density at 60°F.

0.8820 kg/L at 20°C does correspond to 0.8860 relative density at 60°F.

Procedures for Conversion from API Gravity at 60°F to Density at 20°C

1. Convert API to Density at 60°F

$$\rho_{60} = 999.012 \times \frac{141.5}{131.5 + °\text{API}}$$

2. Calculate coefficient of thermal expansion.
   a) Set $\rho_T = \rho_{60}$ rounded to nearest 0.1 kg/m^3.
   b) Determine values of K_0, K_1, and K_2 according to API gravity and applicable table.

Table	API Range		K_0	K_1	K_2
	Min.	Max.			
A	0	100	341.0957	0.0	0.0
B	52	85	192.4571	0.2438	0.0
B	48	52	1489.0670	0.0	−0.00186840
B	37	48	330.3010	0.0	0.0
B	0	37	103.8720	0.2701	0.0
D	−10	45	0.0	0.3488	0.0

   c) Calculate value of α_T according to the formula

$$\alpha_T = \left(K_0 + K_1\rho_T + K_2\rho_T^2\right)/\rho_T^2$$

   d) Round the value of α_T to the nearest 10^{-7}.

3) Calculate the corrected density according to the formula

$$\rho_{20} = \rho_{60}\,(1 - 8\alpha_T)$$

Round the final density to the nearest 0.1 kg/m^3.

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Procedures for Conversion from Relative Density at 60°F/60°F to Density at 20°C

1. Convert Relative Density (60°/60°F), G, to Density at 60°F

$$\rho_{60} = 999.012 \times G$$

2. Calculate coefficient of thermal expansion.
   a) Set $\rho_T = \rho_{60}$ rounded to nearest 0.1 kg/m^3.
   b) Determine values of K_0, K_1, and K_2 according to relative density and applicable table.

Table	Relative Density Range		K_0	K_1	K_2
	Min.	Max.			
A	0.6110	1.0760	341.0957	0.0	0.0
B	0.6530	0.7705	192.4571	0.2438	0.0
B	0.7710	0.7885	1489.0670	0.0	−0.00186840
B	0.7890	0.8395	330.3010	0.0	0.0
B	0.8400	1.0750	103.8720	0.2701	0.0
D	0.8000	1.1640	0.0	0.3488	0.0

   c) Calculate value of α_T according to the formula

$$\alpha_T = \left(K_0 + K_1\rho_T + K_2\rho_T^2\right)/\rho_T^2$$

   d) Round the value of α_T to the nearest 10^{-7}.

3) Calculate the corrected density according to the formula

$$\rho_{20} = \rho_{60}\,(1 - 8\alpha_T)$$

Round the final density to the nearest 0.1 kg/m^3.

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API TITLE*1993 INT ** ■ 0732290 0534193 729 ■

Issue Date: November 17, 1993

Publication: *Manual of Petroleum Measurement Standards*

Edition/Date: General Question

Question: This writer requested a technical opinion regarding the deduction of sediment from a particular cargo.

Reply: This issue is not specifcally addressed in the American Petroleum Institute's *Manual of Petroleum Measurement Standards*. Since this situation is not covered in the scope of the measurement standards, API is not in a position to offer a technical opinion on the issue, particularly as a resolution to a commercial dispute.

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API TITLE*1993 INT ** ■ 0732290 0534194 665 ■

Issue Date: March 4, 1993

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: With respect to 3.3.1, Multiple Qualification—General, what is the recommended direction of travel for welding the branch-on-pipe connection?.

Reply 1: API Standard 1104 does not recommend a direction of welding. The direction of welding is to be the same as was used when qualifying the welding procedure. Please note in 3.3.2 that if the direction of welding is changed in the procedure specification, the welder, using the new procedure, must be requalified.

Question 2: With respect to 3.3.1, Multiple Qualification—General, if the recommended practice is to weld the seam in sectors, what shall the sector arc size be, and in what direction shall the weld be laid within the arc-segment?

Reply 2:. API Standard 1104 does not address your question. As stated in Reply 1, welding must be in accordance with the qualified procedure.

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API TITLE*1993 INT ** ■ 0732290 0534195 5T1 ■

Issue Date: March 4, 1993

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: Background: Company is changing name and consolidating their subsidiary companies and joint ventures which are under their direct control. These organizations have QUALIFIED WELDING PROCEDURES (WPS) under their various names. Is it permissible to use the new company name on all previously qualified WPS's without requalification and documentation? If additional documentation is required, to what extent?

Reply 1: API does not state who must qualify a welding procedure. Therefore, if you have a welding procedure that has been established and qualified per 2.1, you need not requalify the procedure. Please see 2.4.1 which states "Changes other than those given in 2.4.2 may be made in the procedure without the need for requalification, provided the procedure specification is revised to show the change." Since the company name is not listed in 2.4.2, changing the company name does not require requalification of the procedure.

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API TITLE*1993 INT ** ■ 0732290 0534196 438 ■

Issue Date: March 10, 1993

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: In response to your request for information on how to fill out the Coupon Test Report Form that is in API Standard 1104.

Reply 1: The form that is in API Standard 1104 is only a suggested form and you may develop your own form if desired.

The term "Location" normally refers to a municipality, like a city, and "State" refers to one of the 50 states in the United States. You could revise the form to include location terms that are better suited for your needs.

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API TITLE*1993 INT ** ■ 0732290 0534197 374 ■

Issue Date: October 4, 1993

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question 1: A welding procedure for joining 12-inch, X-46 pipe has previously been qualified. If the base materials of X-52 and/or X-60 pipe grades are compatible with the X-46 pipe, can these two pipe grades be incorporated into the procedure without the necessity of additional weld procedure qualification testing?

Reply 1: No. Please refer to 2.3.2.2, Pipe and fitting materials. Here it states that materials may be grouped provided that the qualification test is made on the material with the highest specified yield strength in the group. In your case, the procedure test was made on X-46 pipe so it would not apply to higher grades.

Question 2: Does a change in amperage and voltage in a welding procedure constitute an essential variable?

Reply 2: For manual welding, API Standard 1104 does not consider a change in amperage and/or voltage to be an essential variable.

Question 3: Does API Standard 1104 have any guidelines for preheating or postheating pipe for welding procedures on different grades of pipe material?

Reply 3: No.

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API TITLE*1993 INT ** 0732290 0534198 200

Issue Date: October 7, 1993

Publication: API Standard 1104, *Welding of Pipelines and Related Facilities*

Edition/Date: Seventeenth Edition, September 1988

Question: Refer to 2.6 of API Standard 1104, Table 2 requires up to 8 side-bend test specimens for weld procedure qualification tests on pipes having wall thickness greater than ½ inch. For procedure tests on pipes having wall thickness less than or equal to ½ inch, no side bend test specimens are required. Shall the side bend test be required if pipes having wall thickness greater than ½ inch are used due to availability of material for locations where the design conditions indicate that the required wall thickness based on strength is far less than ½ inch?

Reply: API Standard 1104 is not a design standard. Its purpose is to present methods for the production of high quality welds through the use of qualified welders using approved welding procedures, materials and equipment. Its purpose is also to present methods for production of high quality radiographs to ensure the proper analysis of welding quality through the use of qualified technicians, approved methods and equipment. Therefore, the requirements of Table 2 apply to the wall thicknesses required by the design of the final product. The reason for side bend specimens for wall thickness greater than ½ inch is due to the fact that the clearance in the bend test fixture will not accommodate the root and face bend specimen.

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API TITLE*1993 INT ** ■ 0732290 0534199 147 ■

Issue Date: September 17, 1993

Publication: API 510, *Pressure Vessel Inspection Code*

Edition/Date: Seventh Edition, March 1992

Question: After experiencing a reduction in wall thickness of up to 1.5 millimeters (3 millimeters corrosion allowance), is it in order to conduct a pressure test at 10.5 $kg/cm^2(g)$?

Reply: There is no requirement in API 510 for pressure testing a vessel after loss of half the corrosion allowance. Therefore, it is an owner/user option.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534200 799 ■

Issue Date: September 17, 1993

Publication: API 510, *Pressure Vessel Inspection Code*

Edition/Date: Seventh Edition, March 1992

Question: Is it the intent that the pressure test required for rerating per 5.3.d of API 510 be conducted per 4.4 at a pressure in accordance with the code used for determining the maximum allowable working pressure for the new service conditions of design temperature and pressure?

Reply: Yes, your interpretation is correct. However, 5.3.d also allows the use of special non-destructive testing in lieu of pressure testing.

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API TITLE*1993 INT ** 0732290 0534201 625

Issue Date: September 17, 1993

Publication: API 510, *Pressure Vessel Inspection Code*

Edition/Date: Seventh Edition, March 1992.

Question 1: Under API 510, is the shell of a heat exchanger considered to be a pressure vessel and, therefore, in need of inspection according to the interval set forth in 4.3?

Reply 1: Yes. Heat exchanger shells constructed to a pressure vessel code are considered pressure vessels.

Question 2: Does the bundle of an exchanger fall under any aspect of API 510?

Reply 2: Heat exchanger bundles are considered to be part of the pressure vessel inspection to the extent that they contain pressure retaining members that could release their contents to the surrounding environment, if they were to fail. API 510 does not specifically cover the need for inspection of the tubulars.

Question 3: Must a tube bundle be removed from the shell of an exchanger to facilitate an internal visual inspection that satisfies the requirements of API 510?

Reply 3: API 510 does not currently address the need for removal of the exchanger bundle from the shell for inspection purposes. If it is determined that an internal inspection of the exchanger shell is required, then it would be necessary to remove the exchanger bundle to provide access to the shell. On the other hand, if circumstances are such that an on-stream inspection will satisfy the inspection interval requirements, then it may not be necessary to remove the exchanger bundle.

Question 4: Can the inspection requirements of API 510 be satisfied for a heat exchanger by solely using ultrasonic thickness measurements of the heads and shell?

Reply 4: Yes. On-stream inspection with non-destructive inspection can be used to satisfy inspection interval requirements, under appropriate circumstances. Those circumstances are being clarified in a supplement to API 510, 4.3, that will be issued in 1993.

Question 5: Under API 510, can an exchanger inspection be waived by management if they deem the exchanger to be in non-hazardous, non-critical service?

Reply 5: According to API 510, all pressure vessels subject to corrosion shall be periodically inspected at intervals according to 4.3. If exchangers are in non-corrosive service and meet the four requirements of 4.3 for non-corrosive service, then internal inspection is unnecessary. Otherwise, exchanger shells, according to API 510, shall be inspected at half-life or ten years, whichever is less. However, under the appropriate circumstances, this inspection may be on-stream.

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API TITLE*1993 INT ** 0732290 0534202 561

Question 6: Can an inspector document that he has performed an acceptable inspection of a heat exchanger, in compliance with API 510, Section 4, if the tube bundle remains in the shell; the condition of the vessel is ascertained by looking at the tubes through the shell nozzles and by looking at the tube IDs; and, by visually inspecting the shell with the channel box removed?

Reply 6: Yes, under certain circumstances, an acceptable inspection of an exchanger can be conducted without bundle removal. Those circumstances should be determined by engineering and/or inspection personnel or organizations that are technically qualified to maintain and inspect pressure vessels.

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API TITLE*1993 INT ** 0732290 0534203 4T8

Issue Date: September 17, 1993

Publication: API 510, *Pressure Vessel Inspection Code*

Edition/Date: Seventh Edition, March 1992.

Question 1: Does the term *any straight line* include lines drawn along any plane, that is, circumferential, longitudinal, or some angle?

Reply 1: Yes.

Question 2: If any one of the three provisions are exceeded but not all, does this mean the pitting cannot be ignored?

Reply 2: Yes.

Question 3: If such pitting is not to be ignored, how is it to be evaluated for repair?

Reply 3: Repairs shall be made in accordance with API 510, Section 5. API Recommended Practice 572, *Inspection of Pressure Vessels*, First Edition, February 1992, also gives some guidance on repair techniques.

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API TITLE*1993 INT ** ■ 0732290 0534204 334 ■

Issue Date: January 5, 1993

Publication: API Recommended Practice 520, *Sizing, Selection, Installation of Pressure-Relieving Devices in Refineries,* Part 1, "Sizing and Selection"

API Standard 2000, *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*

Edition/Date: Recommended Practice 520, Fifth Edition, November 1988
Standard 2000, Fourth Edition, September 1992

Question 1: Is this interpretation acceptable?

In API Standard 2000 a composite heat input curve is given in Tables 2 and 3 and Appendix B for three types of tanks. This curve becomes, for medium pressure (<15 pounds per square inch gauge, >1 pound per square inch gauge) and refrigerated tanks above 2800 square feet area, the API Recommended Practice 520 recommendation given by Equation D-2.

For low pressure tanks (<1 pound per square inch gauge) the same composite curve is followed to 2800 square feet area, but above this area, no further increase in heat input is expected.

Reply 1: Yes. Your interpretation is acceptable. Although an environment factor for drainage and fire-fighting effort is not specifically mentioned in Table D-3 of API Recommended Practice 520, the net effect is as indicated in your inquiry. Users may elect to use Equation D-2, which reflects good drainage, and then apply an environment factor from Table D-3 as appropriate.

Question 2: Does the API Standard 2000 composite curve assume "adequate drainage and fire-fighting equipment" when applied to these three types of tanks? If not, how is the heat input above 2800 square feet reconcile with the API Recommended Practice 520 equation which assumes drainage?

In API Standard 2000 environment factors are given in 1.3.2.7 for non-refrigerated tanks and in Table 4 for refrigerated tanks. These factors are consistent with each other and with the factors given in API Recommended Practice 520 (Table D-3) except that 1.3.2.7 introduces a "factor of 0.5 when drainage away from the vessel or tank is provided" for non-refrigerated tanks.

Reply 2: No. The equations and tables in API Standard 2000 do not assume adequate drainage and fire-fighting. A reduction in required vent capacity may be taken as indicated in 1.3.2.7 for non-refrigerated tanks when adequate drainage away from the tank is provided. Additional credit for drainage is not now allowed for refrigerated tanks. However, for refrigerated tanks, and for tanks with design pressures exceeding 1 pounds per square inch gauge, and with greater than 2800 square feet wetted surface area, the equation used in Figure B-1 and for Table 4 is the same as pre-

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API TITLE*1993 INT ** 0732290 0534205 270

sented in API Recommended Practice 520,which includes consideration for drainage. At the time that the third edition of API Standard 2000 was issued, this equation was the only equation for heat input presented in API Recommended Practice 520.

Question 3: If the API Standard 2000 composite curve allows for "adequate drainage," under what conditions can this environment factor of 0.5 for drainage be used, and can it also be applied to refrigerated tanks and API Recommended Practice 520 tanks? The total rate of emergency venting (API Standard 2000) or total heat absorption (API Recommended Practice 520) "shall be multiplied by the environmental factor." The singular case is always used. In all the environment factor tables, only one factor is physically possible, except for non-refrigerated tanks in API Standard 2000, 1.3.2.7, where both drainage and insulation factors could be used, but again the singular case is used.

Reply 3: The factor of drainage may be applied to non-refrigerated tanks wherever provisions are made in the plant design and layout to convey spilled flammable materials away in a manner to substantially reduce the extent of fire exposure to the tank. API Standard 2000 does not now allow the application of a drainage environmental factor to refrigerated tanks. API Recommended Practice 520 does not allow application of an environmental factor for drainage since drainage is implied in Equation D-2.

Question 4: Is it acceptable to multiply heat input/vent rate by more than one environment factor where physically possible?

I am sure that you are aware of some minor misprints in API Recommended Practice 520, as follows:

a. In Figure 26, the horizontal axis values for Specific Heat ratio are a factor 10 too high—the decimal point is missing.

b. In Table D-3, Note b should read—
"...thermal conductivity of 4 BTU/hr–Ft^2–°F per inch at 1600°F..."

c. In Figure E-2, the formula on the graph should read—

$$C = 520 / \{n(2/(n+1))^{[(n+1)/(n-1)]}\}.$$

Reply 4: No. Standard 2000 and API Recommended Practice 520 do not allow application of more than one environment factor. However, as indicated in the response to Question No. 1, Equation D-2 in API Recommended Practice 520 currently allows application of an environment factor to an Equation D-2 that implicitly reflects provision for drainage.

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API TITLE*1993 INT ** ■ 0732290 0534206 107 ■

Issue Date: April 30, 1993

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question: A discrepancy has arisen concerning F.1.2.1. Could you please confirm that any of the components, and thus, the resultant, may exceed the range specified in Table 2 (on page 10) by no more than a factor of 2.

Reply: Recognizing the potential for offsetting loads, F.1.2.1 allows the individual force and moment components to be increased to a maximum of twice the values allowed in Table 2. However, the resultant applied forces and moments must meet the requirements of F.1.2.2.

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API TITLE*1993 INT ** ■ 0732290 0534207 043 ■

Issue Date: April 30, 1993

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question 1: Do constant level oilers that have oil level set by adjusting the position of the flow tube within the adapter by use of a locknut, and the reservoir assembly is a slide fit onto the oiler body and is pushed against a shoulder each time it is installed meet the intention of 2.9.2.2?

Reply 1: Yes.

Question 2: Is there a separate API standard for constant level oilers?

Reply 2: API does not have a separate standard for constant level oilers.

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API TITLE*1993 INT ** 0732290 0534208 T8T

Issue Date: April 30, 1993

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question: What is the justification for the difference between allowable unbalance per 2.8.4.4 and that given by ISO 1940?

Answer: API does not provide underlying reasons for requirements in its standards. API believes the requirements in 2.8.4.4 of API Standard 610 are achievable.

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API TITLE*1993 INT ** ■ 0732290 0534209 916 ■

Issue Date: April 30, 1993

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question 1: Paragraph 2.4 (page 9), External Nozzle Forces and Moments, suggest that Figures 1–5 (pages 10–11) be revised to indicate direction of forces and moments rather than as vectors (using right-hand rule) as currently shown.

Reply 1: The moments called for in Table 2 are to be applied to each pump nozzle using the right-hand for sign convention (see Appendix F). The forces are to be applied as vectors with the positive sign convention as shown by the direction of the arrows in Figure 1.5. The eighth edition will attempt to clarify both the figures and Appendix F

Question 2: Paragraph 3.3.1.6 (page 29), second sentence of the paragraph, seems in conflict with Table 8 (page 29); suggest that letter *Y* and *Z* be reversed in the paragraph.

Reply 2: Combined moment MY_c results in a displacement in the *Z* direction; combined moment MZ_c results in a displacement in the Y direction. This is indicated in Table 8. The clarification issue will be addressed in the eighth edition.

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API TITLE*1993 INT ** ■ 0732290 0534210 638 ■

Issue Date: April 30, 1993

Publication: API Standard 610, *Centrifugal Pumps for General Refinery Service*

Edition/Date: Seventh Edition, February 1989

Question: Does 2.8.4.4 of API Standard 610 contain an error? Why is the maximum allowable residual unbalance so unrealistically low?

Reply: There is no error in 2.8.4.4. Residual unbalance levels of 4W/N or I 6350 W/N are achievable. The dividends of an increase in MBTF that this level of residual unbalance will achieve are significant in a refinery application.

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API TITLE*1993 INT ** ■ 0732290 0534211 574 ■

Issue Date: January 5, 1993

Publication: API Standard 614, *Lubrication, Shaft-Sealing, and Control-Oil Systems for Special-Purpose Applications*

Edition/Date: Second Edition, January 1984

Question 1: The inquiry suggests a conflict of requirements between certain paragraphs. Some paragraphs require stainless steel reservoir, appendages, and piping; other paragraphs require that certain components of the oil system be made of steel.

Reply 1: API Standard 614 covers the minimum requirements for oil systems. The intent of API Standard 614 is to require the use of stainless steel wherever possible for oil-wetted surfaces, especially for parts which are difficult to clean or keep free of rust. Carbon steel is permitted for certain components, and this is spelled out precisely. There is no contradiction; if stainless steel is not specifically required, then it does not have to be provided. If the standard allows the use of carbon steel for certain components, or specifically requires only steel to be used, then the manufacturer is not obligated to provide stainless steel.

Question 2: API Standard 614 does not mention materials of construction downstream of the oil filter, while equipment standards such as, API Standards 616, 618 and 672 clearly mandate stainless steel piping and tubing downstream of the oil filters.

Reply 2: API Standard 614 covers only the console (that is, total oil supply system) which serves the equipment. The purchaser of the console specifies and supplies the interconnecting piping between the console and the equipment (reference 2.13.1). All piping within the confines of the console shall be stainless steel (reference 2.13.1 and 2.13.12 of the second edition), unless otherwise specified.

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API TITLE*1993 INT ** 0732290 0534212 400

Issue Date: January 5, 1993

Publication: API Standard 617, *Centrifugal Compressors for General Refinery Service*

Edition/Date: Fifth Edition, April 1988

Question: In 2.7.3.4 of API Standard 617, the factor 95,540 in the SI Unit does not correspond to 63,000 in the Imperial Units assuming the 63,000 is correct, the factor should be 9546, for the SI Unit Formula applicable to all materials listed in ASME B16.34?

Reply: We agree. We will round to 9550. This will be addressed in the next edition of the standard.

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API TITLE*1993 INT ** ■ 0732290 0534213 347 ■

Issue Date: January 5, 1993

Publication: API Standard 617, *Centrifugal Compressors for General Refinery Service*

Edition/Date: Fifth Edition, April 1988

Questions:

Does the impact requirements in 2.11.3 of API Standard 617 apply to weldments and heat affected zone in addition to steels?

Does the impact requirements in 2.11.3 of API Standard 617 apply to weldments and heat filler materials also?

If impact requirements are applicable to steels only, then why is ASME VIII, Division I, UG 84 specified?

Does criteria for ASME VII UG 84(a) general paragraph considered to be applicable to compressors by referring to ASME VIII, UG 84? Which paragraphs of ASME VIII UG 84 are applicable to compressors purchased per API Standard 617?

Does the statement "...Impact strength sufficient to qualify..." mean that only steels shall have a sufficient impact strength, or steels and its weldments and heat affected zones shall have sufficient impact strength?

Reply: Unless API Standard 617 makes a specific exception to reference steel only and not welds, reference to Section VIII, Division I, UG 84 of the ASME Code means base metal, weld metal and heat affected zone.

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API TITLE*1993 INT ** 0732290 0534214 283

Issue Date: April 30, 1993

Publication: API Standard 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tank*

Edition/Date: Eighth Edition, June 1990

Question 1: Does API Standard 620 provide rules for the design of frangible roof joints?

Reply 1: No.

Question 2: Does API Standard 620 provide rules and procedures for operating tanks?

Reply 2: No.

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API TITLE*1993 INT ** ■ 0732290 0534215 11T ■

Issue Date: April 30, 1993

Publication: API Standard 620, *Design and Construction of Large, Welded, Low-Pressure Storage Tank*

Edition/Date: Eighth Edition, June 1990

Question: Is it the intent in Q.6.3.3 of API Standard 620 to require charpy impact testing of HAZ specimens for austenitic stainless steel welds for service temperatures below -200°F?

Reply: No.

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API TITLE*1993 INT ** ■ 0732290 0534216 056 ■

Issue Date: April 30, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question 1: In API Standard 650, do the rules of 3.7.3 take precedence over Figure 3-4A for weld spacing?

Reply 1: Yes. Figure 3-4A is being revised to eliminate any confusion.

Question 2: Does API Standard 650 provide rules for spacing with riveted tank apparatus?

Reply 2: No. API Standard 650 is titled, *Welded Steel Tanks for Oil Storage.*

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API TITLE*1993 INT ** 0732290 0534217 T92

Issue Date: April 30, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question: In F.1.2 of API Standard 650, is the *nominal* weight of the metal that counterbalances the internal pressure calculated in the uncorroded condition?

Reply: Yes, unless otherwise specified by the purchaser.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534218 929

Issue Date: April 30, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question: Concerning the use of low-hydrogen electrodes in API Standard 650, does 5.2.1.10 override 2.8.1?

Reply: Yes.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534219 865 ■

Issue Date: April 30, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question: Does API Standard 650 require only spot radiography for a joint efficiency of 1.0?

Reply: API Standard 650 in the basic document does not use the concept of joint efficiency inthe thickness formulas. In Appendix A, the designer has a choice of radiography requirements (spot or none) and is required to apply an associated joint efficiency (.85 or .70).

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534220 587 ■

Issue Date: April 30, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question: Are there any circumstances which would allow the use of API Standard 650 within the industry for the re-erection of a tank?

Reply: No. API Standard 650 covers only construction. API Standard 653 covers re-erection of carbon and low-alloy aboveground steel storage tanks which are nonrefrigerated and designed for atmospheric pressure.

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API TITLE*1993 INT ** ■ 0732290 0534221 413 ■

Issue Date: November 5, 1993

Publication: API Standard 650, *Welded Steel Tanks for Oil Storage*

Edition/Date: Eighth Edition, November 1988

Question: Is a hydrostatic test of a tank that has undegone major repairs or alterations as defined in 10.3.1.2 of API Standard 653 required in addition to the testing of welds in accordance with 10.1.5, 10.1.6, and 10.1.7?

Reply: A tank that has undergone a major repair or alteration, such as removal or replacement of the shell-to-bottom weld, must have a hydrostatic test unless exempted by the conditions given in 10.3.2. Since your letter did not state whether you could satisfy the conditions of 10.3.2, the tank should have a hydrostatic test.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534222 35T

Issue Date: January 14, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question 1: Referring to 2.3.2.1 of API Standard 653, First Edition, is L required to be placed somewhere on t_2, even if a lower t_1 can be found above or below t_2, or even if t_2 is not in the vertical plane containing the critical length L?

Reply 1: No.

Question 2: Referring to 2.3.2.2 of API Standard 653, First Edition, is the *minimum acceptable tank shell thickness* equal to the thickness defined as t_{min} in 2.3.3?

Reply 2: Yes.

Question 3: What should be done to analyze an area of corrosion that extends over a weld and/or over two tank courses of different thicknesses?

Reply 3: API Standard 653 does not specifically address this question, which will be referred to the appropriate subcommittee for study. Paragraph 2.3.3.5 permits design by analysis, which could be applied to this situation.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534223 296 ■

Issue Date: April 30, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question 1: Does API Standard 653 require partial removal of lining (>0.05 inch thick and installed in accordance with API Recommended Practice 652) on a tank bottom as part of a complete internal inspection to measure the bottom plate thickness if continuous effective cathodic protection has been provided and a complete visual inspection and spark test of the lining reveal no problems?

Reply 1: API Standard 653 requires that bottom plate thickness measurements be taken. The need to remove a reinforced lining to measure bottom plate thickness is not covered by API Standard 653.

Question 2: May thickness credit be taken for an adhesive-applied steel patch on a tank bottom plate which has an applied tank bottom reinforced lining, >0.05 inch thick in accordance with API Recommended Practice 652 if it has been confirmed that the adhesive is compatible with the stored product?

Reply 2: No. Non-welded patch plates are not acceptable details, so no thickness credit can be taken against the minimum required bottom plate thickness.

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API TITLE*1993 INT ** ■ 0732290 0534224 122 ■

Issue Date: July 19, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: In API Standard 653, does Figure 10-1 override the restrictions of Table 10-1, or is the figure an additional requirement?

Reply: For a hydrotest exemption to be permitted, Figure 10-1 must be satisfied for tanks having materials of unknown toughness. Refer to Basic Condition 1 in Table 10-1.

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API TITLE*1993 INT ** ■ 0732290 0534225 069 ■

Issue Date: July 19, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: When installing an annual ring to API Standard 653, whether an annual ring is required by API Standard 650 or not, do the x-ray requirements per 6.1.2.10 of API Standard 650 apply?

Reply: Paragraph 7.1.1 of API Standard 653 requires radiographic examination per 6.1.2.10 of API Standard 650 for the annular plate welds when the annular plates are required by 3.5.1 of API Standard 650.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534226 TT5 ■

Issue Date: July 19, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: If a TK311 stainless steel tank which was constructed, installed, and used in one state is later cut apart, moved to another state and re-erected in the second state, which API standards within the industry apply to the cutting and subsequent re-erection in the second state?

Reply: None. There is presently no standard for the re-erection of a stainless steel tank.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534227 931 ■

Issue Date: July 20, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: Does API Standard 653 allow fillet-welded patch plates on the shell?

Reply: No. We are considering a revision to API Standard 653 that will establish the parameters under which fillet-welded patch plates will be allowed on the shell.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534228 878

Issue Date: July 20, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: Refer to 10.2.3.2 of API Standard 653, should the term *work* be interpreted to mean *welds*?

Reply: Your question has been referred to the appropriate subcommittee for further study; and your recommendation for a revision of the standard is being pursued.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534229 704 ■

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question 1: Does API Standard 653 specify qualification requirements for the person performing diesel oil tests on shell-to-bottom welds per 10.1.6?

Reply 1: No.

Question 2: Does 10.1.6 of API Standard 653 require that a vacuum-box test and a diesel oil test both be performed on the completed shell-to-bottom?

Reply 2: No. Either test is acceptable. The first pass must be tested with light diesel oil per 10.1.6.

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API TITLE*1993 INT ** 0732290 0534230 426

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: Is it the intent of API Standard 653 that, in addition to conducting the inspection of a tank, the Certified Inspector is also responsible for interpretation of results, approval to continue usage of the tank, making specific recommendation for repairs (for example, weld type, weld size, and material), and certification of a reconstructed tank?

Reply: The precise roles and duties, beyond those defined in Section 4, of the Certified Inspector are not defined. We plan to revise the standard to address this issue.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534231 362 ■

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question 1: Is each person performing field inspection of aboveground storage tanks required to be a Certified Inspector per API Standard 653 or could technicians who do not hold certificates perform data gathering and report the findings to a Certified Inspector who would then make evaluations and recommendations?

Reply 1: The precise roles and duties beyond those defined in Section 4 of the Certified Inspector are not defined. We plan to revise the standard to address this issue.

Question 2: Could a company have a single person certified who could evaluate data gathered by technicians?

Reply 2: The precise roles and duties, beyond those defined in Section 4, of the Certified Inspector are not defined. We plan to revise the standard to address this issue.

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API TITLE*1993 INT ** 0732290 0534232 2T9

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Questions: Does 10.3 of API Standard 653 require hydrotesting for the following:

1. A repair on an internal floating roof?
2. A complete replacement?
3. Adding a manway to the shell, designed per API Standard 650?
4. Shell plate removal and replacement to accomplish 1 or 2?

Replies:

1. No.
2. No, but H.7.3 of API Standard 650 requires a flotation test for this new internal roof.
3. Yes, assuming the manway is below the design liquid level. However, if the conditions for exemption of hydrostatic testing are met (per 10.3.2), hydrotesting is not required.
4. If the shell plate that was removed included more than 12 inches of vertical weld, then a hydrotest is required, unless the exemption permitted by 10.3.2 is exercised, and all of the required conditions are met. If the shell plate that was removed did not include more than 12 inches of existing vertical weld (door sheet), then a hydrotest is not required, as long as all conditions in 10.3.2 are met (see 10.3.2.3).

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534233 135 ■

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: Does 7.13.3.1 of API Standard 653, quoted below, apply to the welds of reinforcing plates?

7.13.3.1 Minimum spacing in any direction (toe-to-toe of welds) between the hot tap and adjacent nozzles shall be equivalent to the square root of RT–where R is the tank shell radius, in inches, and T is the shell plate thickness, in inches.

Reply: Yes.

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API TITLE*1993 INT ** ■ 0732290 0534234 071 ■

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Question: In API Standard 653 is 6 inches the minimum diagnostic length of each radiograph?

Reply: Yes, in accordance with 10.2.1.6 of API Standard 653.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534235 T08 ■

Issue Date: September 17, 1993

Publication: API Standard 653, *Tank Inspection, Repair, Alteration, and Reconstruction*

Edition/Date: First Edition, January 1991, Incorporates Supplement 1, 1992

Questions: The following questions refer to 9.1.1 of API Standard 653 for a tank with unknown or obsolete steel material:

1. May a weld repair be made using a weld procedure previously qualified on P1 to P1 material?
2. Must the weld procedure and welder be requalified using the actual plate from the tank?
3. Is the verification of weldability met by a successful bend test of a coupon from the actual plate to be welded, if the coupon was welded using the previously qualified P1 to P1 weld procedure and welder?

Replies:

1. The weld procedure must comply with the current rules of ASME Section IX.
2. No.
3. Yes.

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API TITLE*1993 INT ** 0732290 0534236 944

Issue Date: April 30, 1993

Publication: API Standard 671, *Special-Purpose Couplings for Refinery Service*

Edition/Date: Second Edition, February 1990

Question: The question is two-fold, does the piloting requirement for intermediate and high speed coupling of 2.1.10.1 include the flexible element? If yes, then can the bolts become the piloting mechanism?

Reply: The answer for the first question is yes. The answer for the second is no, the bolts cannot act as the piloting mechanism (see 2.1.11.2). The reason is that the coupling design shall be such that balance can readily be achieved and maintained. The key word is *maintained.* Bolts which are removable for maintenance are not considered to provide a long term pilot.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534237 880 ■

Issue Date: August 2, 1993

Publication: API Standard 671, *Special-Purpose Couplings for Refinery Service*

Edition/Date: Second Edition, February 1990

Question: Your question concerns the possible interpretation of 2.1.1 of API Standard 671, Second Edition, and competitive suppliers catalog ratings. The question centers on the true service factor of a catalog ratings relative to the API specification.

Reply: The intent in 2.1.1 of API Standard 671, , is to mandate an adequate experience factor to allow for off-design operation which may occur at operating points requiring higher torque than the normal operating point on which the coupling selection is based, as well as equipment variation resulting in higher torque than actual equipment design point.

An experience factor of 1.57 is proposed, however, the accompanying note allows decrease of this factor to 1.25 if rotor dynamic considerations mandate a lower overhung moment than can be otherwise reasonably achieved.

The base paragraph also asks the vendor to furnish couplings with the maximum practical service factor. The experience factor of 1.75 which is applied in addition to the service factor is a guideline; not an absolute.

If a machinery train was only to operate at its normal point, and that normal point met its horsepower and speed exactly, an experience factor of 1.0 would be acceptable. The additional experience factor proposed in 2.1.1 of API Standard 671, is to cover range of operation and variation in equipment design tolerances such as being 4 percent low in efficiency or 5 percent high in head on a compressor, running maximum steam conditions on a turbine, operation in the field at higher than specified densities.

It is expected that a manufacturer will incorporate additional factors to account for material property variations, engineering predictive tolerances, transients, and so forth. The manufacturer's design factors should be applied to the rating to establish a factor of 1.0 at the normal operating point (coupling selection point). The experience factor referred to in 2.1.1 of API Standard 671, must then be applied in addition to properly size the coupling to meet API Standard 671 criteria.

To summarize, coupling manufacturer's design factors which must be applied to establish a base rating for the coupling at the normal operating point, create a base rating which a factor of 1.0 at the normal operating point to which the API experience factor must then be applied. The manufacturer's design factors may not be applied in a manner so as to reduce the API experience factor criteria.

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API TITLE*1993 INT ** ■ 0732290 0534238 717 ■

Issue Date: January 5, 1993

Publication: API Standard 672, *Packaged, Integrally Geared Centrifugal Air Compressors for General Refinery Service*

Edition/Date: Second Edition, April 1988

Question 1: Refer to Figure 1, page 7, of API Standard 672. One pressure switch high (PSH) is shown in the delivery line of the standby pump and it reads "Main pump running alarm." We think it should read "standby pump running alarm."

Reply: We agree with the position that the PSH switch should indicate "standby pump running alarm." This will be corrected in future editions.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534239 653 ■

Issue Date: January 5, 1993

Publication: API Standard 675, *Positive Displacement Pumps-Controlled Volume*

Edition/Date: First Edition, March 1980; Reaffirmed, December 1987

Question 1: What is the difference between the motor nameplate ratings required by the first and second sentences in 4.1.2 of API Standard 675, First Edition?

Reply 1: It is good engineering practice to have the relief valve operate before the driver exceeds its rating. Paragraph 2.8.1 requires the relief valve setting to be at least 10 percent above the rated discharge pressure. Consequently, a motor driver must have a sufficient power rating to operate the pump at the relief valve setting, including 10 percent accumulation, to comply with the second sentence of 4.1.2. This is a higher power than 110 percent of the greatest power for any specified operating point, since the rated flow at the rated discharge pressure is an operating point.

Question 2: What types of drivers are likely to be used with pumps built to API Standard 675?

Reply 2: Paragraph 4.1.1 states that the purchaser will specify the type of driver, implying that many types are used.

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API TITLE*1993 INT ** ■ 0732290 0534240 375 ■

Issue Date: January 5, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: Are Caterpillar 1G2 test results that were obtained prior to March, 1992 valid for licensing SH/CD engine oils?

Reply: No. API Publication 1509, Twelfth Edition, 1.2.7, specifies that "For engine tests that use either or both of API S and API C Service Categories ... the engine tests covered by the CMA code shall be conducted in accordance with the CMA Code." (MTAC does not apply to the 1G2). The CMA Code went into effect on March 30, 1992.

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API TITLE*1993 INT ** ■ 0732290 0534241 201 ■

Issue Date: February 25, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: What is a definition of *complete engine testing* as found on the EOLCS application form, page 39, of API Publication 1509?

Reply: *Complete engine testing* means that the oil has passed each engine test required for licensing of that oil (see Appendix G-1, page 33). Where applicable, (for all "S" category oils) the oil must have been registered with the Chemical Manufacturers Association (CMA) monitoring agency (Registration Systems, Inc.) and have passed each test in full compliance with the CMA Product Approval Code of Practice. When a complete engine test program has been successfully completed on an oil, API Guidelines for SAE Viscosity-grade Engine Testing, see Appendix F, page 27, can be applied for licensing. The intent of complete engine testing is to distinguish between oils which have been fully tested and those that are licensed by reference to *read-across*.

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API TITLE*1993 INT ** ■ 0732290 0534242 148 ■

Issue Date: February 25, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: On the EOLCS application, Product Traceability Code, page 43 of API Publication 1509, is the day and time of manufacture actually required to be submitted as part of the EOLCS application for licensure?

Reply: The Introduction to Part D, Product Traceability Code, page 43 of API Publication 1509 states in part "API is mandating, as a requirement of licensing, that each container of licensed oil marketed be **legibly date stamped** (bolding added) and that sufficient information be provided in licensing documents to allow API to interpret the date stamp and match the audited oil with the licensing data for that oil."

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534243 084

Issue Date: February 26, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: What is an acceptable API ILSAC mark design display on containers?

Reply: Page 2 of API Publication 1509, Twelfth Edition, shows how the mark must appear on labels; 4.1.2.1 specifies design requirements for the mark. Paragraph 4.1.2.1 states in part, "The background of the **outer band** (bold added) ... shall be a color that contrasts with the label background."

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534244 T10

Issue Date: March 8, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: A marketer of motor oil, for example, a car manufacturer, sells oil under its own name. The marketer purchases its branded oil from several different oil companies, each of which use a completely different format for its product traceability code. How should this be handled under EOLCS?

Reply: Each separate formulation supplied to the marketer must be filed with and accepted by API on a Part B form, Technical and Program Data Sheet. Each Part B has space for five (5) separate formulations. If a marketer has more than 5 formulations an additional Part B form must be submitted in the licensing document.

API must be able to compare data obtained from oils analyzed under the aftermarket audit program with the data for that oil submitted to API as part of the licensing program. Therefore, an interpretation of the Formulation Code identifier required on Part B for each oil must be available to API. The marketer should submit a separate Part D form, Product Traceability Code, for each formulation and link the information in Part D with Part B.

Please refer to the Introduction to Part D, Product Traceability Code, on page 43 of API Publication 1509 which states in part "... API is mandating, as a requirement of licensing, that each container of licensed oil marketed be **legibly date stamped** (bolding added) and that sufficient information be provided in the licensing documents to allow API to interpret the date stamp and match the audited oil with the licensing data for that oil."

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534245 957 ■

Issue Date: March 31, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: With regard to the administration fee, does one fee cover the various oils listed on the Application Form? With regard to the volume fee, is this based on the total sales of the various oils on the Application Form? (It is assumed that the volume of each grade is not separately assessed).

Reply: There are two fees as described in 3.2 of API Publication 1509, Twelfth Edition. The administration fee is a flat fee payable by all licensees ($500 in 1994). It covers all licensed oils. The volume of sales fee covers the total sales of licensed oils, that is, the *oils listed on the Application Form*. The volume of sales fee is not separately assessed.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534246 893 ■

Issue Date: May 17, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Questions: On the EOLCS application, Product Traceability Code, on page 43 of API Publication 1509, does the "date stamp" requirement refer to

1. The date of manufacturer or packaging?
2. Is the **actual day** (bold added) of either manufacturer or packaging required?

Replies:

1. You must use date of packaging for the date stamp.
2. The day of packaging is required.

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API TITLE*1993 INT ** ■ 0732290 0534247 72T ■

Issue Date: May 21, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: Assume there are two Group 1 base oils from different manufacturers, both of which have passed engine and bench tests with **given** (bolding added) additive and VI package and both individual base oil/additive blends are licensed "SH."

Could a packager purchase these two different soured based oil, **mix them together** (bolding added) with the given additive/VI package and have an API licensed motor oil? This assumes (1) the SAE viscosity grade of the theoretical blend was one licensed for each base oil, and (2) the additive/VI packages and treat levels of the two licensed formulations were consistent.

Reply: Please refer to Appendix E, page 21, of API Publication 1509, Twelfth Edition, and the **bolded** words in the question.

Given is undefined in your question. Is the same additive package used in both base oils or is the additive package different? For this question we assume the same additive package has been used for both base oils tested and that both have been fully tested in accord with Appendix E.

Mix them together is not defined and this is the crucial point. According to E.2.1.3, "Engine testing is not required when a single interchange base stock that meets either the Group 1 or Group II designation is used at less than or equal to 10 mass percent of the blended PCMO." If the tested PCMO oils are both in Group I, **and** have the same additive package blended into the other at less than or equal to 10 mass percent then testing is not required. If however, the blending of one with the other is at greater than 10 mass percent then the instructions in Table E-2 must be followed, namely the new blend must pass two engine tests—Sequence IIIE and Sequence VE.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534248 666 ■

Issue Date: May 27, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Questions: Appendix E, 2.2.2, is as follows:

Complete performance documentation is required for the original PCMOs. The detergent inhibitor (DI) and/or viscosity modifier (VM) remain unchanged when interchange base oils are tested, except as provided by **the CMA Code** (bolding added). A base oil interchange obtained under these guidelines applies to a single PCMO formulation. In the event of a change in the DI and/or VM outside of the CMA Code, these guidelines shall be **reapplied**. (bolding added)

What part of **the CMA Code** is referred to?

How should the guidelines (BOIG) be **reapplied**?

Reply: In **the CMA Code** proposed changes in the *core data set* are accomplished by applying Appendix H, Guidelines for Minor Formulation Modifications. Proposed changes in a *program* are accomplished by applying Appendix I, Program Guidelines.

The BOIG should be reapplied as follows. If the VM/DI concentration in the interchange base oil differs from the original PCMO oil by more than the variation authorized under the CMA Code (Appendix H or I), then complete performance documentation is required for the new PCMO with the interchange basestock.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534249 5T2 ■

Issue Date: May 27, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: On the EOLCS application, Product Traceability Code, on page 43 of API Publication 1509, is it adequate to indicate only the month and year of manufacture?

Reply: No. The day of manufacture is also required. This response assumes that the date of *manufacture* and *packaging* is the same day.

The Introduction to Part D, Product Traceability Code, on page 43 of API Publication 1509 states in part "... API is mandating, as a requirement of licensing, that each container of licensed oil marketed be **legibly date stamped** (bolding added) and that sufficient information be provided in the licensing documents to allow API to interpret the date stamp and match the audited oil with the licensing data for that oil.

The date that appears on the container that is selected for aftermarket auditing must be the date of packaging.

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API TITLE*1993 INT ** ■ 0732290 0534250 214 ■

Issue Date: June 18, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question 1: Part B of the EOLCS Application for Licensure specifies a test method for measuring sulfur and nitrogen which is not addressed in the text of API Publication 1509. Are licensees bound by requirements specified in the Application for Licensure?

Reply 1: Yes, the API License Agreement requires licensees to comply with all requirements specified in API Public;ation 1509 and the Application for Licensure.

Question 2: If test methods for measuring physical and chemical properties are specfied in API Publication 1509, may marketers utilize alternate (but equivalent) test methods to measure these properties?

Reply 2: No. The properties must be measured by the methods specified in API Publication 1509.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534251 150 ■

Issue Date: June 30, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: In Appendix G, Requirements for API Service Category SH by Viscosity Grade, page 33, of API Publication 1509, Twelfth Edition, the last column titled "All Other Grades" has an NR for all Bench Tests and the L-38 shear stability test. NR means No Requirement?

Reply: The technical language which describes API Service Category SH (2.3.2.1, page 2, API Publication 1509) specifically mandates the application of specified parts of ILSAC GF-1 or DOD CID A-A-52039 specifications for all viscosity grades covered by these specifications. This includes the three oils specifically noted in Appendix G (SAE 5W-30, 10W-30 and 15W-40) **and** any other oils which are requested to be licensed under ILSAC GF-1. Also, the most recent edition of SAE J300 contains high temperature/high shear requirements for SH.

Appendix G, Table G-1, as it appears in API Publication 1509, Twelfth Edition, is correct with the addition of the SAE J300 requirements as noted above. If ASTM subsequently adopts requirements for SH, these requirements will be presented to the appropriate committees for consideration as an amendment to API Publication 1509.

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API TITLE*1993 INT ** ■ 0732290 0534252 097 ■

Issue Date: January 5, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: If an oil company sells its oil to another company and that company resells the oil under its own brand name does the reseller have to be licensed?

Reply: Yes. Paragraph 1.2.1 of API Publication 1509, Twelfth Edition, includes the following statements. EOLCS is "... a licensing system that includes a formal license agreement executed by the marketer with API." An oil marketer "is defined as the marketing organization responsible for the integrity of the brand name and the representation of the product in the marketplace."

Section 8 of the EOLCS License Agreement is as follows:

> 8. Licensee agrees that it is the marketing organization responsible for the integrity of the brand name and the product's representation in the marketplace **and agrees to use the marks only on products bearing the Licensee's name** (bolding added).

The licensees name on the product is the key element. If the owner company's name appears on the container it may license the oil. If the subsidiary's name appears on the container, the subsidiary must be separately licensed.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534253 T23 ■

Issue Date: July 12, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: When is the last date that API Service Category SG can be displayed alone in the API Service Symbol (donut).

Reply: API will not license API Service Category SG **alone** after December 31, 1993 and licensees can display the donut **only** on product manufactured **before** December 31, 1993.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534254 96T ■

Issue Date: October 26, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: May Energy Conserving or Energy Conserving II be used in the API Service Symbol *donut* with SG in 1994?

Reply: Paragraph 2.3.2.2 prescribes the language for SG. The language includes the option to include Energy Conserving II in the symbol if the oils meet the requirements of DOD CID A-A-52039 and ILSAC GF-1.

Paragraph 2.3.2.3 prescribes the Energy Conserving Classification. Selected sentences from 2.3.2.3.1 are quoted as follows:

"The energy conserving oil classification for gasoline – powered passenger cars, vans... may be used only in conjunction with API Service Category SH and SGT. ... Testing for conformance to these categories must be in accordance with the CMA Code."

If a marketer wishes to display an Energy Conserving classification with SGT, the marketer must have the oil tested under the CMA Code. The original testing program (Sequence VI) to establish Energy Conserving, conducted when the original SGT testing program was run, is not valid to claim Energy Conserving in 1994.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534255 8T6 ■

Issue Date: October 28, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: May CD be used in the API Service Symbol *donut* alone or with SG in 1994?

Reply: Paragraph 2.3.1 was modified effective September 1, 1993 as follows: "During 1994, Service Category SG may be used only with CF-4 and/or CE and/or CD-II, and when displayed in the Symbol, the C Category must appear first."

The crucial words in 2.3.1 are "... SG may be used **only** (bold added) with ..." This means that CD may not be used in the *donut* with SG. However, CD may be used by itself or with any of the other heavy duty categories.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534256 732

Issue Date: February 5, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: A foreign oil company (API licensed) wishes to keep confidential the additive and VI improver packages it is providing to one of its customers. The customer is selling the oil under its name (as a *rebrand*) and will be licensed by API. The oil company proposes, as an alternative to providing all the information to the customer, to provide only a proprietary code for the additive and VI improver packages—which will mask the name of the additive manufacturer. Thus, the customer will receive the CMA candidate data package with coded data. Can the customer be licensed by API under these circumstances?

Reply: Yes, the customer can be licensed.

What API does require for licensing is a "YES" attestation in the block related to the line "CMA Petroleum Additives Panel Product Approval Code of Practice" on Part B of the application form (see page 39 of API Publication 1509, Twelfth Edition. (At the present time, this "YES" attestation is required for "S" category and ILSAC licensed oils only.)

As an API licensee, the customer can receive from API an inquiry about any oil which it has licensed. It is possible that as a result of a monitoring or enforcement inquiry API will request of the licensee certain information which might include, for example, information on the VI Improver. In that case the licensee would be required to provide to API, as specified in the license agreement, all necessary information to satisfy the monitoring or enforcement request.

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API TITLE*1993 INT ** ■ 0732290 0534257 679 ■

Issue Date: March 31, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: The API Application for Licensure forms includes spaces for several oil brands and viscosities. It is assumed that alternate brand names for the same oil (and the same viscosity) are acceptable for inclusion on this form.

Reply: Each product, defined as a separate brand or viscosity grade, requires a separate PART B in the Application for Licensure package. The purpose of this requirement is to insure that oils selected from the market for analysis in API's aftermarket audit program can be identified and matched with the data presented for licensing. For example, if the same oil (SAE viscosity grade) is packaged under the label XYZ as well as another label ABC, they are separate oils for purposes of licensing, should be submitted as separate candidates for licensing.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** 0732290 0534258 505

Issue Date: March 31, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: Once the Application Form has been submitted and the license obtained, is it possible to change or modify a brand name? Or does a new Application Form and a further administration fee need to be paid?

Reply: If a licensee desires to change or modify a brand name for a specific product, a new Part B must be completed and submitted to API for approval. No additional fees are required.

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API TITLE*1993 INT ** 0732290 0534259 441

Issue Date: January 5, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: API licenses two marks. What is the correct way to refer to these marks in which are identified in API Publication 1509, Twelfth Edition as "... the API Service Symbol and the ILSAC Certification Mark" (2.1). Our company wishes to advertise these marks to both our retail and bulk customers. However, "ILSAC" and "Service Symbol" have no meaning for most customers. What other labelling or identifying terminology are we permitted to use when referring to the marks?

Reply: Section 2.1 specifies the official names of the two marks.

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API TITLE*1993 INT ** 0732290 0534260 163

Issue Date: June 30, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: We wish to license a product but to market it under more than one name. For example, we plan to market a single formulation in a number of different geographical locations. How should we proceed?

Reply: The EOLCS aftermarket audit program will match the data obtained from an analysis of marketplace samples with the data submitted to API for licensing, for the same formulation. The brand name is one of the elements required to make the match. Therefore, the name of each brand must be available to API. You should submit a separate Part B for each brand.

This interpretation was issued as a result of a question on the above edition. It applies to all previous and subsequent editions in which the text remains substantially unchanged.

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API TITLE*1993 INT ** ■ 0732290 0534261 0TT ■

Issue Date: November 15, 1993

Publication: API Publication 1509, *Engine Oil Licensing and Certification System*

Edition/Date: Twelfth Edition, January 1993

Question: May the words *Energy Conserving* or *Energy Conserving II* or *API Service* be translated into a foreign language within the API Service Symbol (donut).

Reply: No.

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